

Review

Phase Change Materials for Photovoltaic Thermal Management: A Comprehensive Review of Material Innovations and Hybrid Architectures

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Abstract

The escalating global demand for renewable energy has positioned solar photovoltaics (PV) as a critical technology for achieving net-zero emissions. However, PV efficiency is strictly limited by thermal degradation, where elevated operating temperatures significantly reduce power output and accelerate material aging. This review systematically evaluates the integration of advanced phase change materials (PCMs) as a passive thermal management solution. We analyze the transition from material-level innovations—including nano-enhanced PCMs, 3D conductive frameworks, and shape-stabilization—to system-level hybrid architectures such as liquid—PCM, heat pipe-fin, and thermoelectric generator (TEG) integrations. Synthesis of recent empirical data (2024–2026) demonstrates that optimized PCM composites can achieve PV temperature reductions of up to 32 °C and electrical efficiency enhancements exceeding 19%. Furthermore, techno-economic assessments reveal that these systems can reduce the levelized cost of energy (LCOE) by 5–15% and achieve energy payback times as short as 1.5 years. Finally, this paper identifies critical research gaps in long-term outdoor durability, AI-driven predictive modeling, and sustainable bio-based encapsulation, providing a strategic roadmap for the commercialization of next-generation solar thermal management systems.

Keywords: photovoltaic thermal management; passive cooling; phase change material

1. Introduction

1.1. The Global Photovoltaic Context and the Net-Zero Imperative

The transition toward a decarbonized global energy infrastructure is heavily predicated on the rapid scaling of renewable technologies, with solar photovoltaics (PV) assuming the most critical role. In strict alignment with the net-zero emissions (NZE) Scenario, which outlines a stringent pathway for limiting global temperature rise to 1.5 °C, the expansion of solar PV infrastructure must achieve unprecedented milestones [1,2]. Contemporary projections establish that actualizing the NZE Scenario necessitates an annual global solar PV generation level of approximately 9200 TWh by 2030, representing a massive increase from the current baseline of 1600 TWh [1]. To manifest this reality, the global PV market must sustain an aggressive annual capacity addition, racing toward an accumulated 11,500 GW target [3]. As massive installations increasingly dominate hyper-arid and high-irradiance zones to maximize solar yield, the severe operational challenges induced by extreme ambient temperatures have become a focal point of global research [4].



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1.2. The Thermal Penalty in PV Systems

While solar PV technology effectively harvests incident photons, its operational efficiency is intrinsically constrained by a severe “thermal penalty.” When exposed to high solar irradiance, elevated operating temperatures precipitate a cascading degradation of both electrical efficiency and long-term material integrity [5]. For conventional monocrystalline and polycrystalline silicon modules, the temperature coefficient ranges from -0.4% to -0.5% per $1\text{ }^{\circ}\text{C}$ increase above the standard test condition (STC) of $25\text{ }^{\circ}\text{C}$. Emerging next-generation technologies, such as perovskite solar cells, exhibit an even more acute thermal sensitivity, rendering thermal management a critical requirement for their commercial viability [6].

Beyond immediate power loss, chronic thermal stress significantly accelerates material degradation within the module architecture. Sustained operating temperatures exceeding $60\text{ }^{\circ}\text{C}$ induce severe thermochemical breakdown of the Ethylene Vinyl Acetate (EVA) encapsulant. This leads to EVA “yellowing,” which significantly attenuates optical transmittance and promotes severe delamination at the glass-encapsulant interface [7–9]. Furthermore, severe thermal cycling exacerbates micro-cracking in the silicon wafers and accelerates the degradation of solder joints, radically shortening the operational lifespan and increasing the levelized cost of energy (LCOE) [10,11].

1.3. Evolution of PV Cooling Technologies and the Emergence of PCMs

To counteract these formidable thermal barriers, various cooling paradigms have been developed. Active cooling techniques, such as forced water circulation or air convection, exhibit robust heat removal capacities but have fundamental limitations: they require continuous parasitic power (e.g., pumps and fans), suffer from high maintenance complexities, and involve high capital expenditures [12,13]. Conversely, phase change materials (PCMs) have emerged as the widely researched passive thermal buffering solution. Operating as zero-energy thermodynamic sinks, PCMs leverage their high latent heat of fusion to absorb transient peak thermal loads, maintaining the PV module in an isothermal state without consuming electrical power or requiring moving parts [14,15].

1.4. Review Methodology and Novelty

To ensure a rigorous and transparent analysis, this review adopted a systematic literature search strategy. Primary databases, including Web of Science, Scopus, and IEEE Xplore, were utilized, applying Boolean keyword combinations such as (“photovoltaic” OR “PV”) AND (“phase change material” OR “PCM” OR “thermal management”) AND (“nano-enhanced” OR “hybrid PV/T”). The screening procedure prioritized peer-reviewed articles published between 2020 and 2026, excluding non-peer-reviewed reports to ensure data reliability. While several review articles on PV-PCM systems exist, they predominantly focus on specific architectural integrations (e.g., building-integrated PVs) or isolated active cooling methods. The novelty of this manuscript lies in its comprehensive bridging of material-level micro-innovations (such as MXene/TiO₂ and 3D porous frameworks) with macro-level hybrid system architectures (TEG and liquid-PCM), evaluated strictly through a recent techno-economic and lifecycle assessment (LCA) lens. Table 1 summarizes the specific contributions of this review compared to previously published reviews, highlighting our unique focus on recent empirical data (2024–2026) and levelized cost of energy (LCOE) optimizations. To provide a highly analytical framework, this review is explicitly organized around the primary design trade-offs governing PV cooling: thermal conductivity versus latent heat capacity (Section 3), leakage control versus added parasitic mass (Section 3.3), and passive simplicity versus hybrid system complexity (Section 4). This

structure aims to guide researchers in evaluating and designing highly specific PV thermal management strategies.

2. Mechanisms of Heat Generation and Thermal Regulation in PV-PCM Systems

2.1. Physics of PV Heat Generation

The operational architecture of commercial PV systems is fundamentally governed by the quantum mechanics of semiconductor light absorption and charge carrier transport. While silicon-based PV cells operate near a theoretical power conversion efficiency ceiling of approximately 20% to 25%, the vast majority of the incident solar spectrum—often exceeding 70%—is absorbed as heat rather than being converted into useful electrical work [16,17]. The accumulation of this thermal energy is primarily driven by two microscopic mechanisms:

- **Thermalization Losses:** Photons possessing energy greater than the semiconductor's bandgap ($E > E_g$) excite electrons into high-energy states within the conduction band. These “hot carriers” rapidly relax to the conduction band edge through phonon emission, converting the excess energy ($E - E_g$) entirely into thermal vibrations within the crystalline lattice [18,19].
- **Non-Radiative Recombination:** Charge carriers that fail to be extracted often undergo non-radiative recombination at defect sites or impurities. This process releases energy as phonons instead of photons, further elevating the module temperature and significantly degrading the open-circuit voltage (V_{oc}) [20,21].

The macroscopic energy balance for a PV module can be defined as [17,22]:

$$G_{tot} = P_{el} + \rho_{pv} \cdot G_{tot} + q_{loss} \quad (1)$$

where G_{tot} is the incident solar radiation, P_{el} is the electrical power output, ρ_{pv} is the reflectance, and q_{loss} represents the total thermal loss, which must be actively managed to maintain system stability.

2.2. Thermodynamics of PCM Thermal Regulation

When a PCM layer is structurally integrated onto the rear backsheet of a PV module, it creates an intelligent, self-actuating thermodynamic buffer [23]. As the solar irradiance increases, the PCM initially absorbs heat through its sensible heat capacity, leading to a linear rise in temperature. While the fundamental thermodynamic principles of this solid–liquid transition are governed by universal thermal energy storage mechanisms [21], the remainder of this review explicitly prioritizes PV-specific empirical evidence to avoid unsupported analogies, as their specific application to PV modules requires unique geometric and irradiance considerations [24]. During this isothermal process, the PCM absorbs a substantial amount of energy (latent heat of fusion) without a significant increase in temperature, effectively “clamping” the PV module temperature during peak sun hours [22]. As ambient temperatures drop below T_m during nighttime, the PCM undergoes an exothermic solidification process, rejecting the stored heat and resetting its latent capacity for the next diurnal cycle [25].

2.3. Fundamental Bottlenecks of Pure PCMs in PV Applications

Despite the theoretical benefits, the practical deployment of pristine organic PCMs in PV systems faces three critical bottlenecks that necessitate the development of advanced composites [21,26]:

- Extremely low thermal conductivity: Most organic PCMs exhibit thermal conductivities below $0.3 \text{ W/(m}\cdot\text{K)}$, which prevents rapid heat absorption from the PV cells and creates a high interfacial thermal resistance.
- Thermal saturation (the thermal blanket effect): If the solar heat load exceeds the PCM's latent heat capacity, the material becomes fully melted (saturated). In this state, the liquid PCM acts as a thermal insulator, trapping heat and exacerbating thermal degradation.
- Inclinal leakage and containment failure: Terrestrial solar installations are typically tilted. When pure PCMs melt, the combination of hydrostatic pressure and volumetric expansion often leads to liquid leakage, formation of insulating air voids, and potential fire hazards.

3. Material-Level Innovations: Advanced PCM Composites for PV Cooling

To overcome the inherent limitations of pure phase change materials (PCMs)—specifically their intrinsically low thermal conductivity and severe susceptibility to liquid leakage at elevated temperatures—material-level enhancement strategies have become the core focus for optimizing photovoltaic (PV) thermal management. This section systematically reviews recent advancements in enhancing PCM performance through nanoparticle doping, the construction of three-dimensional (3D) porous frameworks, and advanced encapsulation technologies tailored for solar applications. A comprehensive comparison of the cooling performance and efficiency enhancements achieved by these various material-level strategies is summarized in Table 1.

3.1. Nano-Enhanced Phase Change Materials (NPCMs) for PV Applications

The introduction of nanoscale conductive fillers can effectively reduce the thermal response time of PCMs, which is crucial for managing the highly transient nature of solar irradiation and sudden heat fluxes in PV modules.

- High-conductivity nanofillers and empirical performance: The incorporation of high-thermal-conductivity nanomaterials, such as graphene, carbon nanotubes (CNTs), and 2D MXene, significantly improves the phonon transport efficiency within the composite matrix [27–29]. Recent empirical studies from 2024 to 2026 demonstrate the profound impact of these nanocomposites in real-world PV applications. For instance, the application of graphene-enhanced PCMs has been proven to reduce PV surface temperatures by $10.04 \text{ }^{\circ}\text{C}$, yielding a 12.15% boost in electrical efficiency [30]. Similarly, hybrid graphene–CNT formulations achieved an $11 \text{ }^{\circ}\text{C}$ reduction combined with a 14.3% electrical efficiency enhancement [31], while the application of cutting-edge PCM nano-emulsions as novel coolants has been experimentally demonstrated to significantly enhance heat extraction, elevating the system's maximum thermal-equivalent energy efficiency to 89.23% [32].
- Dispersion stability and functionalization: A critical challenge for NPCMs is the agglomeration of nanoparticles during repeated melting-solidification cycles. Studies have demonstrated that the surface functionalization of nanomaterials (e.g., functionalized CNTs) can effectively mitigate particle aggregation, ensuring consistent thermal conductivity and long-term thermophysical stability over extended operational periods [33].
- Eutectic and photothermal PCMs: By utilizing eutectic phase change materials [34] or integrating fillers with excellent photothermal properties [35,36], the composites can not only precisely tune the melting temperature to match the PV operating environment but also enhance the overall solar energy utilization and electro-thermal conversion efficiency.

3.2. Hierarchical 3D Conductive Frameworks and Porous Matrices

While several advanced 3D matrix designs and highly conductive composites were initially pioneered in Battery Thermal Management (BTM) systems, their fundamental thermodynamic principles—specifically mitigating thermal hotspots and accelerating heat dissipation—are highly transferable and increasingly adopted in PV cooling research. However, to ensure PV-specific relevance, the quantitative evidence synthesized in the subsequent discussions relies strictly on data directly demonstrated in photovoltaic cooling applications rather than extrapolations from BTM.

For large-scale PV modules, relying solely on nanofillers is often insufficient to achieve macroscopic temperature uniformity. The integration of 3D continuous conductive frameworks provides a more efficient and structurally robust heat transfer pathway across the entire panel area [37].

- Expanded graphite, metal foams, and quantitative gains: Utilizing the worm-like porous structure of expanded graphite (EG) or the highly interconnected ligaments of metal foams (e.g., copper or aluminum foams) constructs a highly integrated thermal network [38–40]. Quantitative validations of these 3D frameworks reveal substantial performance gains. PV modules integrated with copper foam-PCM composites have recorded temperature reductions of 14.5 °C and electrical efficiency improvements of 16.2% [41]. Concurrently, expanded graphite (EG) composites have demonstrated a 12.8 °C cooling effect, translating to a 15.4% increase in power output [42].
- Hotspot mitigation and mismatch losses: In PV thermal management, 3D conductive skeletons play a vital role in eliminating localized hotspots. Maintaining temperature homogeneity across the solar array is fundamentally important to minimize electrical mismatch losses, which can severely degrade the overall power output of the system [43].
- The thermal-mass trade-off: While 3D metal frameworks significantly enhance heat dissipation, their introduction adds considerable parasitic weight to the system. This thermal-mass trade-off, influenced by the porosity and pore size of the metal foam, must be carefully balanced against the load-bearing capacity of the PV mounting structures [44].

3.3. Advanced Shape-Stabilized PCMs (SSPCMs) and Inclination Leakage Prevention

Unlike horizontally flat systems, solar panels are almost exclusively installed at inclined angles in outdoor environments. This architectural necessity makes the prevention of liquid leakage during the phase transition the most critical design challenge for PCM-based PV cooling.

Advanced shape-stabilized PCMs (SSPCMs) address this issue by encapsulating the phase-change components within polymer, melamine foam, or multi-structured graphite matrices [45–47]. Remarkably, even when the active PCM reaches its fully molten state, the composite retains its macroscopic solid form, entirely eliminating the risk of leakage, structural contamination, or system failure on inclined PV surfaces.

Enclosing PCMs within micrometer-scale shells (e.g., GO/SiO₂ hybrid shells or crosslinked polystyrene) further increases the specific surface area and heat exchange efficiency [48,49]. Recent advancements in microencapsulated PCMs (e.g., using melamine/acrylic shells) not only reduced PV temperatures by 9.5 °C and boosted efficiency by 11.2%, but also robustly restricted leakage rates to less than 1% [50].

For utility-scale solar farms, macro-encapsulation using 3D-printed structures, spherical macrocapsules, or robust metallic containers provides higher cost-effectiveness, ease of installation, and mechanical durability [51,52]. Notably, modern macro-encapsulated polymeric PCM matrices have achieved a 13.2 °C temperature reduction with a 14.8%

efficiency boost, while demonstrating a negligible leakage rate (<1%) even under high inclination angles [53]. To synthesize these material-level advancements, Table 2 provides a critical comparative analysis of the fundamental advantages, limitations, and commercial viability of different PCM architectures.

Table 1. Performance comparison of advanced PCM composites for PV cooling.

PCM Category	Enhancement Material	Study Type & Configuration	Operating Conditions	Temp. Reduction (°C)	Efficiency Boost (%)	Citations
Nano-Enhanced	Graphene	Experimental (Indoor)	1000 W/m ² , ambient 25 °C	10.04	12.15%	[30]
	Hybrid Graphene-CNT	Experimental (Outdoor)	Peak summer irradiance	11	14.3%	[31]
	PCM Nano-emulsions	Experimental (Indoor)	1000 W/m ² , active flow	18	19.5%	[32]
3D Framework	Copper Foam	Experimental (Outdoor)	Natural convection	14.5	16.2%	[41]
	Expanded Graphite (EG)	Experimental (Indoor)	1000 W/m ²	12.8	15.4%	[42]
Shape-Stabilized	Melamine/Acrylic Shell	Numerical & Exp.	800 W/m ² , inclined setup	9.5	11.2%	[50]
	Polymeric Matrix	Numerical Optimization	Transient thermal load	13.2	14.8%	[53]

Table 2. Critical comparative analysis of different PCM architectures for PV thermal management.

PCM Architecture	Thermal Conductivity (W/mK)	Leakage Risk	Relative Cost	Key Advantages	Practical Limitations
Pure Organic PCMs	Low (<0.3)	High	Low	High latent heat capacity	Thermal saturation, severe leakage.
Nano-Enhanced PCMs	Moderate (0.5–2.0)	Moderate	High	Fast thermal response	Nanoparticle agglomeration.
3D Frameworks (Metal Foam)	High (2.0–10.0)	Low	High	Excellent temperature uniformity	High parasitic weight, corrosion.
Shape-Stabilized PCMs	Moderate (0.5–1.5)	Negligible	Moderate	No leakage on inclined PVs	Reduced latent heat capacity.

It is critical to note that direct quantitative comparison of the performance metrics presented in Table 1 and throughout this review is inherently limited by the heterogeneity of the literature. The reported temperature reductions and efficiency boosts are derived from vastly different experimental boundaries, including variations in climate, solar irradiance (e.g., indoor simulators vs. outdoor transient conditions), PCM mass, encapsulation designs, and PV cell technologies. Consequently, these values should be interpreted as contextual indicators of potential rather than standardized, one-to-one comparative benchmarks.

4. Hybrid PCM-Based Thermal Management Architectures for Photovoltaics

4.1. The Necessity of Hybrid Systems: Overcoming Thermal Saturation

While phase change materials (PCMs) provide significant passive heat-buffering capabilities, their efficacy is inherently constrained by the “thermal saturation” phenomenon. In high-irradiation environments or prolonged solar exposure, the low intrinsic thermal conductivity of organic PCMs often leads to localized melting and a subsequent loss of cooling capacity. Once fully melted, the PCM layer can even act as a “thermal blanket,” trapping heat within the PV module and causing temperatures to soar beyond optimal

efficiency thresholds. Consequently, hybridizing PCMs with active cooling (water), secondary conductive pathways (fins/heat pipes), or waste-heat recovery systems (TEGs) is essential to dynamically regenerate the PCM's latent capacity and maintain long-term thermal symmetry [54]. The quantitative performance metrics and distinctive features of these hybrid architectures are detailed in Table 2 for comparative analysis.

4.2. Liquid-PCM Hybrid PV/T Systems

The integration of active liquid cooling with passive PCM buffers (forming PV/T-PCM systems) is widely considered the most effective strategy for managing extreme thermal loads. In this configuration, the PCM acts as a transient buffer that absorbs peak heat fluxes during midday solar spikes, while the integrated water channels continuously extract the accumulated bulk heat. Recent empirical data (2024–2026) strongly validates this approach: liquid-PCM hybrid systems have been shown to reduce PV cell temperatures by 14.5 to 27 °C compared to bare modules [55]. This thermal regulation translates to an electrical efficiency enhancement ranging from 1.6% up to 22.2% [56]. Furthermore, by capturing the waste heat in the circulating fluid for domestic or industrial heating, the combined electro-thermal efficiency of the system can reach up to 73.47%, with annual total energy efficiencies peaking at approximately 83% under specific boundary conditions (e.g., peak solar irradiance of 1000 W/m² and optimized mass flow rates). It should be noted that this combined efficiency highly depends on the utilization rate of the recovered low-grade waste heat [55,57]. Furthermore, achieving such high metrics in real-world, dynamic outdoor environments remains highly challenging due to fluctuating ambient conditions, thermal contact resistances, and variable wind speeds, which are often perfectly controlled in laboratory settings.

4.3. PCM Hybrids with Heat Pipes and Conductive Fins

For compact or off-grid PV installations where active water pumping is unfeasible, coupling PCMs with phase-change heat pipes or metallic fins provides a robust passive or semi-active cooling solution. Embedded high-conductivity elements serve as “thermal bridges” that rapidly transport heat from the PV-PCM interface deep into the PCM matrix. Recent advancements utilizing highly optimized fin geometries (such as Y-shaped or fractal designs) have achieved significant cooling performance, demonstrating maximum temperature reductions of 26 to 30 °C [58,59]. Field studies indicate that this accelerated heat extraction boosts electrical efficiency by 3% to 15% and can increase the daily total power output by more than 35% [60]. Additionally, integrating nanoparticles with metal fins has been proven to increase the thermal storage capacity by up to 71.6%, ensuring deep and uniform utilization of the PCM's latent heat without reaching premature saturation [59,61].

4.4. PV-TEG-PCM Hybrid Systems

A highly innovative approach to PV thermal management involves the integration of thermoelectric generators (TEGs). In this architecture, the PCM is strategically placed to absorb the PV module's waste heat, acting as a stable, isothermal heat source for the hot side of the TEG. This setup not only cools the PV panel but also scavenges the waste heat to generate supplementary electricity via the Seebeck effect. Recent investigations demonstrate that PV-TEG-PCM systems can effectively reduce module temperatures by up to 30 °C, leading to combined efficiency improvements of 1% to 14% [62,63]. While the discrete electrical contribution from the TEG is modest—adding approximately 1.7% to the annual energy yield—the synergistic seasonal efficiency improvements can reach up to 14.3% during peak summer conditions, making this a viable avenue for total solar energy harvesting [64,65].

5. Economic Feasibility and Lifecycle Assessment (LCA)

While the thermodynamic and electrical benefits of PCM-based PV thermal management are well-established, the large-scale commercialization of these systems is ultimately dictated by their economic viability and environmental sustainability. This section evaluates the critical cost–benefit trade-offs and lifecycle impacts of integrating advanced PCMs into utility-scale solar infrastructure.

5.1. Economic Metrics (EPBT, LCOE, and ROI)

The integration of advanced PCMs fundamentally introduces an initial Capital Expenditure (CAPEX) penalty to the PV module installation. However, recent quantitative evaluations demonstrate that this initial investment is rapidly offset by the enhanced electricity yield and extended operational lifespan of the cooled modules. Contemporary studies (2024–2026) report that optimized PV-PCM systems can achieve a levelized cost of energy (LCOE) reduction of 5% to 15% compared to conventional bare PV modules, particularly in high-irradiance, arid climates [66,67]. These techno-economic projections are contingent upon specific boundary conditions, including an assumed discount rate of 5–8%, a 20-year system lifespan, and deployment in high-irradiance geographical zones. Future cost-performance optimizations must balance the high CAPEX of advanced nanomaterials with long-term maintenance costs. Furthermore, the Energy Payback Time (EPBT)—a critical metric for renewable investments—is significantly shortened. Advanced hybrid systems exhibit an EPBT ranging from 1.5 to 5.1 years, representing a payback period up to 17.5% shorter than their uncooled counterparts [68,69]. Depending on the specific PCM configuration and local electricity tariffs, the return on investment (ROI) period typically spans between 1.05 and 8 years, confirming strong economic viability for long-term deployments. However, these optimistic economic projections largely stem from idealized assumptions. Actual financial viability in uncontrolled outdoor deployments will likely experience longer payback periods due to accelerated material degradation, unpredictable weather anomalies, and unforeseen maintenance overheads.

5.2. Lifecycle Assessment (LCA) and Environmental Impact

Evaluating the true sustainability of PV-PCM systems requires a rigorous lifecycle assessment (LCA) to balance the embodied carbon generated during PCM synthesis against the operational emission savings. On the operational front, PCM-cooled PV systems exhibit quantifiable environmental benefits, achieving a 10% to 30% reduction in total Greenhouse Gas (GHG) emissions (saving up to 4.5 tons of CO₂ annually for utility-scale setups) due to improved electrical efficiency and reduced degradation [70]. Consequently, the payback time for the embodied carbon is remarkably swift, typically ranging from 0.8 to 2.5 years [66,71]. However, a critical environmental bottleneck remains in the encapsulation matrix. While the PCMs themselves (especially inorganic salt hydrates) may possess low global warming potential (GWP), the massive quantities of smelted aluminum or steel required to structurally encapsulate the fluid matrix often generate over 50% of the entire system's embodied carbon footprint [72]. This heavy structural mass also necessitates vastly heavier, high-embodied-carbon mounting infrastructure, posing a challenge to achieving a rapid carbon payback.

5.3. Cost–Benefit Trade-Offs and Commercial Viability

There exists a fundamental cost–benefit dichotomy in the selection of PCMs. Nano-enhanced PCMs (e.g., graphene or MXene-doped) offer high thermal conductivity and peak efficiency; however, their extreme material costs significantly delay the ROI, making them currently less viable for massive utility-scale solar farms [68,70]. Conversely, macro-

encapsulated organic PCMs or salt hydrates integrated with structural polymers can reduce total system costs by 40% to 60% while still delivering adequate thermal buffering [71]. Ultimately, the large-scale commercial success of PV-PCM systems depends not merely on incremental enhancements via expensive nanomaterials, but on profound macroeconomic supply chain innovations. The rapid development of ultra-cheap, lightweight, and low-embodied-carbon macro-encapsulation techniques—potentially utilizing fully recycled structural polymers or advanced biodegradable bio-composites—is the key factor for successfully transitioning this technology into the potential scaling of this technology toward broader utility-level solar applications [72].

6. Conclusions and Future Perspectives

6.1. Summary of Key Research Findings

This review concludes that advanced Phase Change Material (PCM) systems represent a technically feasible approach for photovoltaic thermal management. At the material level, the synergy between nano-scale fillers (such as graphene and MXene) and 3D porous frameworks has partially mitigated the historical limitations of low thermal conductivity, enabling rapid thermal response and isothermal buffering during peak irradiance. At the system level, hybridization with active liquid cooling or thermoelectric generators (TEGs) has pushed total energy harvesting efficiencies to record levels, with some configurations reaching 83% total efficiency. Quantitatively, the adoption of these technologies offers a potential economic incentive, with LCOE reductions up to 15% and carbon payback periods often under 2.5 years, potentially justifying the initial capital expenditure, provided that long-term system durability can be guaranteed.

6.2. Future Perspectives and Research Gaps

Despite the significant progress documented in this review, several critical research gaps remain that must be addressed to facilitate the widespread commercial adoption of PCM-based thermal management systems in PV applications:

- Long-term material durability and outdoor aging: Most current studies focus on short-term indoor thermal cycling (e.g., 100–500 cycles using artificial solar simulators). There is a profound lack of data regarding the physicochemical stability of composite PCMs over the actual lifespan of a commercial PV plant (typically 20–25 years). Research into PCM aging mechanisms, potential sub-cooling effects during long-term seasonal changes, and the impact of continuous harsh outdoor conditions (such as intense UV radiation, high humidity, and wind-induced mechanical vibrations) on encapsulation integrity is urgently needed.
- AI-driven thermal prediction and digital twins: The phase transition front within a PCM matrix is highly non-linear and difficult to monitor in real-time under fluctuating environmental conditions. Future research should leverage Machine Learning (ML) and Digital Twin technology to develop predictive thermal models. By integrating real-time meteorological data (e.g., solar irradiance and ambient temperature forecasts), these models could optimize the “active–passive” switching frequency in hybrid systems. This would allow active cooling components (like water pumps or fans) to trigger preemptively based on predicted latent heat saturation levels, thereby maximizing overall system efficiency.
- Sustainability and circular economy: As the global solar market scales, the environmental footprint of large-scale PCM synthesis and deployment becomes critical. There is a specific research gap in the development of “Bio-based PCMs” derived from agricultural waste and the recyclability of composite PCMs at the end of the solar panel’s

life. Symmetrizing high thermal performance with environmental sustainability will be a key scientific challenge for future PV cooling designs.

- Integration with emerging solar technologies: While this review primarily focuses on traditional silicon-based PV panels, the thermal requirements for emerging technologies, such as perovskite solar cells and bifacial modules, are vastly different. Perovskite materials are notoriously sensitive to thermal stress, which accelerates their degradation. Investigating how advanced PCMs can be adapted to provide the specific, rigorous thermal stability required for these next-generation solar cells represents a nascent and vital research frontier.

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